

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2142

Lakeland Solar Energy Llc 138 kV
September 20, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2142. The Interconnection Customer proposes a 131.7 MW large load (data center) facility interconnecting at the Lakeland Solar Energy Llc 138 kV substation in the Pennsylvania area. The request was received on July 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 423 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,547,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
12Th & Irving to Lakeland Solar Energy Llc 138 kV	Little Chute to Talenenergy Martins Crk Llc Harrisburg 345 kV	106.4	357	48.3
Mountain 500/345 kV TX	Coronet to Ft Humbug Substation 500 kV	100.9	1,467	12.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
12Th & Irving to Lakeland Solar Energy Lc 138 kV	Reconductor line	\$607,000	67% (\$406,000)
Mtn 500-345 kV Auto	Replace transformer with higher-rated unit	\$27,578,000	8% (\$2,141,000)
	Total allocated to project		\$2,547,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.